CONFIDENTIAL — BUSINESS INTELLIGENCE REPORT

Customer Behavior Analysis Report

Retail Consumer Shopping Behavior — Data Analysis & Strategic Insights

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REPORT DATE
21 December 2025

DATASET SIZE
~3,900 Records

STATUS
Completed ✓

TABLE OF CONTENTS

1. Executive Summary 01
2. Project Overview 02
3. Problem Statement & Business Questions 03

4. Methodology & ETL Process 04
5. Data Sources & Quality Assessment 05
6. Key Performance Indicators (KPIs) 06
7. Data Analysis & SQL Findings 07
8. Dashboard Visualizations 08
9. Insights & Interpretation 09
10. Risks & Challenges 10
11. Recommendations 11
12. Action Plan & Next Steps 12
13. Conclusion 13
14. Appendix — SQL Query Reference 14

01 Executive Summary

This report presents a comprehensive Business Intelligence and Data Analysis of customer shopping behavior for a leading retail organization, covering transactional data from a dataset of approximately 3,900 customer records. The analysis was conducted using Python (Pandas) for data cleaning and transformation, Microsoft SQL Server for structured querying and aggregation, and Power BI for interactive dashboard visualization.

The primary objective was to uncover actionable insights into consumer purchasing patterns, demographic contributions, discount behavior, shipping preferences, and subscription engagement — enabling data-driven strategic decisions across marketing, product, and operations functions.

SUMMARY HIGHLIGHTS

Key Achievement: Male customers generated $157,890 in total revenue — more than double the female segment ($75,191) — indicating a dominant customer base that warrants targeted retention strategies.

Key Insight: 863 customers applied discounts yet still exceeded the average purchase value, confirming that strategic discounting drives higher transaction values.

Key Gap: Subscribed and non-subscribed customers share an identical average spends of $59, indicating the subscription program currently delivers no measurable uplift in purchase behavior.

Critical Finding: 80% of the customer base (3,116 of 3,900) are classified as Loyal customers, yet subscription adoption remains low at 27%, representing a significant missed retention opportunity.

02 Project Overview

BACKGROUND

A leading retail company sought to leverage its historical transactional data to gain a deeper understanding of customer shopping behavior. Changes in purchasing patterns across demographic segments, product categories, and fulfilment channels prompted the management team to commission a structured data analysis initiative. This engagement was designed to transform raw transactional records into strategic intelligence capable of informing commercial decision-making.

PROJECT DETAILS

Attribute	Details
Project Name	Customer Behavior Analysis
Business Unit	Retail Operations / Business Intelligence

Attribute	Details
Lead Analyst	Satish Aryal
Reporting Date	21 December 2025
Dataset Size	~3,900 Customer Records
Database	SQL Server — Database: Customer_Behaviour, Table: customer
Tools Used	Python (Pandas), Microsoft SQL Server 2022, Power BI Desktop
Report Status	Completed

03 Problem Statement & Business Questions

BUSINESS PROBLEM

The retail organization identified significant changes in purchasing behavior across demographic groups, product lines, and sales channels. Management required a data-driven framework to understand which factors — including discounting strategies, customer reviews, seasonal trends, shipping preferences, and payment behaviors — drive repeat purchases and long-term customer loyalty.

> **Overarching Business Question:** "How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

MANAGEMENT QUESTIONS ADDRESSED

#	Business Question	Analysis Method
Q1	What is the total revenue generated by male vs. female customers?	SQL Aggregation — GROUP BY Gender
Q2	Which customers used a discount but still exceeded the average purchase amount?	SQL Subquery — Conditional Filter
Q3	Which are the top five products by highest average review rating?	SQL Aggregation — AVG, ORDER BY DESC
Q4	How does average spend compare across Standard vs. Express shipping?	SQL Aggregation — WHERE IN, GROUP BY
Q5	Do subscribed customers spend more than non-subscribed customers?	SQL Aggregation — GROUP BY Subscription Status
Q6	Which five products have the highest discount application rate?	SQL CASE WHEN — Percentage Calculation
Q7	How can customers be segmented into New, Returning, and Loyal tiers?	SQL CTE — CASE WHEN Segmentation
Q8	What are the top three most purchased products within each category?	SQL Window Function — ROW_NUMBER() PARTITION BY
Q9	Are repeat buyers (5+ purchases) more likely to subscribe?	SQL Filter — WHERE previous_purchases > 5
Q10	What is the revenue contribution of each age group?	SQL Aggregation — SUM by Age Group

SCOPE DEFINITION

✓ **In Scope:** All customer records (~3,900), all product categories (Clothing, Accessories, Footwear, Outerwear), all age groups, all shipping types, discount behavior, subscription status, and purchase frequency analysis.

Out of Scope: Real-time transactional streaming, external market benchmarks, supplier/inventory data, and geographic regional breakdowns beyond available location fields.

04 Methodology & ETL Process

A structured Extract-Transform-Load (ETL) pipeline was implemented to ensure data integrity, analytical consistency, and reliable output generation across all phases of this engagement.

1. **Data Ingestion & Null Value Assessment**

   The raw dataset (~3,900 records) was ingested into a Pandas Data Frame via Jupyter Notebook. A null value audit using df. isnull().sum() identified 37 missing values exclusively within the Review Rating column. All other 16 columns returned zero null values, confirming overall strong data completeness.

2. **Statistical Profiling (EDA)**

   Descriptive statistics were generated using df. describe() to establish baseline distributions: Average Age = 44.07, Mean Purchase Amount = $59.76, Mean Review Rating = 3.75, Mean Previous Purchases = 25.35. These baselines informed subsequent segmentation thresholds and anomaly detection.

3. **Data Standardization & Column Normalization**

   All column headers were converted to lowercase with underscores to ensure consistent referencing across Python and SQL environments (df.columns = df.columns.str.strip().str.lower().str.replace(' ','_')). All string values across applicable columns were similarly normalised to lowercase.

4. **Null Value Imputation**

   The 37 missing Review Rating values were imputed using the median rating per product category (grouped median fill via lambda transform), thereby preserving category-specific rating distributions without distorting the overall mean.

5. **Duplicate Column Detection & Removal**

A field equivalence checks confirmed that the discount_applied and promo_code_used columns were 100% identical across all records (validated via . all () comparison returning np. True_). The redundant promo_code_used column was subsequently dropped to eliminate analytical duplication.

6 **Age Group Feature Engineering**

A derived age_group column was created by partitioning the age field into four equal-frequency quantile bins using pd.qcut(df['age'], q=4): young_adult, adult, middle_aged, and senior — enabling demographic segmentation in downstream analysis.

7 **Purchase Frequency Feature Engineering**

A purchase_frequency_days column was derived by mapping the frequency_of_purchases field to its numerical day equivalent (Weekly=7, Fortnightly=14, Monthly=30, Quarterly=90, Bi-Weekly=14, Every 3 Months=90, Annually=365), enabling time-based purchase cadence analysis.

8 **Data Export to SQL Server**

The cleaned and enriched DataFrame was exported to Microsoft SQL Server 2022 (RTM — 16.0.1000.6) via SQLAlchemy and pyodbc using the ODBC Driver 17 for SQL Server. The data was written to database Customer_Behaviour, schema dbo, table customer (df.to_sql with if_exists='replace').

9 **SQL Analysis & Power BI Visualization**

Ten structured SQL queries were executed against the SQL Server database to address all management business questions. Results were subsequently loaded into Power BI Desktop to produce the Customer Behavior Dashboard, incorporating KPI cards, donut charts, bar charts, and interactive slicers.

05 Data Sources & Quality Assessment

DATA INVENTORY

#	Source / Field	Type	Volume	Null Count	Quality
1	Customer ID	Identifier	3,900 records	0	High
2	Age / Age Group	Demographic	3,900 records	0	High
3	Gender	Demographic	3,900 records	0	High
4	Item Purchased / Category	Transactional	3,900 records	0	High
5	Purchase Amount (USD)	Financial	3,900 records	0	High
6	Review Rating	Behavioral	3,900 records	37 (imputed)	Imputed
7	Subscription Status	Engagement	3,900 records	0	High
8	Shipping Type	Operational	3,900 records	0	High
9	Discount Applied / Promo Code	Promotional	3,900 records	0 (duplicate removed)	High
10	Previous Purchases / Frequency	Behavioral	3,900 records	0	High

DATA QUALITY SUMMARY

The dataset demonstrated a high overall quality score, with null values limited exclusively to the Review Rating field (37 records, representing 0.95% of the total dataset). These were successfully imputed using a category-grouped median strategy. One redundant column (promo_code_used) was identified as a full duplicate of discount_applied and removed. Post-cleaning, 100% of the 3,900 records were retained for analysis — no records required deletion.

06 Key Performance Indicators (KPIs)

3,900
TOTAL CUSTOMERS
Full Dataset Analyzed

$59.76
AVERAGE PURCHASE (USD)
Range: $20 – $100

3.75
AVERAGE REVIEW RATING
Out of 5.0 | Min: 2.5

$233,081
TOTAL REVENUE (USD)
Male: $157,890 | Female: $75,191

27%
SUBSCRIPTION RATE
1,053 of 3,900 customers

863
DISCOUNT + ABOVE-AVG SPENDERS
▲ High-Value Discount Users

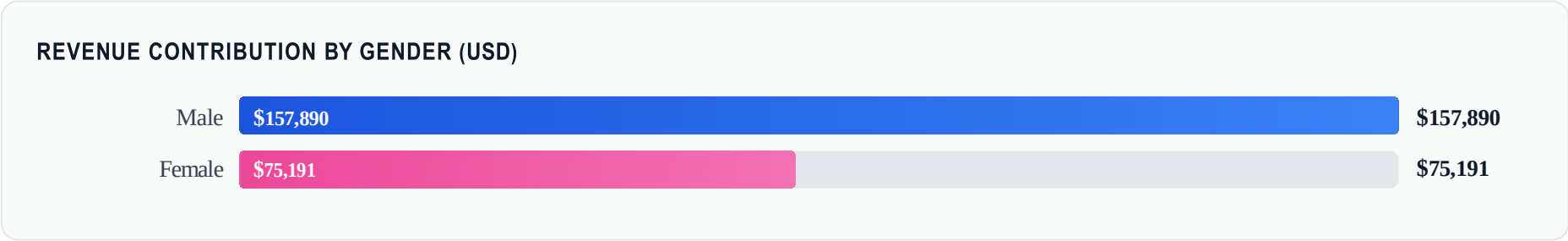
KPI PERFORMANCE SUMMARY

KPI	Value	Segment Detail	Status
Total Revenue — Male	$157,890	68% of total revenue	Strong
Total Revenue — Female	$75,191	32% of total revenue	Growth Opportunity
Subscription Adoption Rate	27%	1,053 subscribers / 2,847 non-subscribers	Underperforming
Loyal Customer Share	80%	3,116 of 3,900 customers	Strong Base
Avg Purchase — Express Shipping	$60	vs. Standard: $58 (+3.4%)	Insight
Repeat Buyers (5+ Purchases)	3,476	958 subscribed / 2,518 not subscribed	Retention Gap

07 Data Analysis & SQL Findings

FINDING 1 — REVENUE BY GENDER

SQL aggregation of total purchase amounts by gender revealed a substantial revenue disparity. Male customers generated **$157,890** compared to **$75,191** from female customers — a difference of $82,699 (110% higher). This suggests that either the product catalogue disproportionately attracts male shoppers or that female customer engagement and acquisition strategies require targeted improvement.



FINDING 2 — DISCOUNT USERS EXCEEDING AVERAGE SPEND

A total of **863 customers** were identified who applied a discount code and yet still spent above the dataset average of $59.76. This finding strongly supports the hypothesis that discounts serve as a transaction trigger that motivates higher overall basket values, rather than simply reducing per-unit revenue. This cohort represents a high-value, price-responsive segment ideal for targeted promotional campaigns.

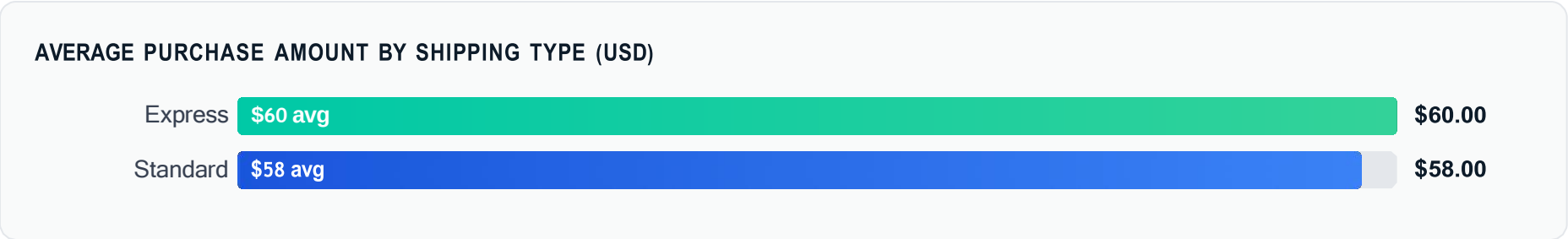
FINDING 3 — TOP 5 PRODUCTS BY AVERAGE REVIEW RATING

Rank	Product	Average Review Rating	Status
1	Gloves	3.86 / 5.00	Top Rated
2	Sandals	3.84 / 5.00	High
3	Boots	3.82 / 5.00	High

Rank	Product	Average Review Rating	Status
4	Hat	3.80 / 5.00	High
5	Handbag	3.78 / 5.00	Above Avg

FINDING 4 — AVERAGE SPEND BY SHIPPING TYPE

Customers opting for Express Shipping recorded a marginally higher average purchase of $60 versus $58 for Standard Shipping. While the differential is modest, it suggests that customers willing to pay a premium for faster fulfilment also demonstrate a marginally higher willingness to spend — a pattern consistent with higher purchase intent at the point of checkout.

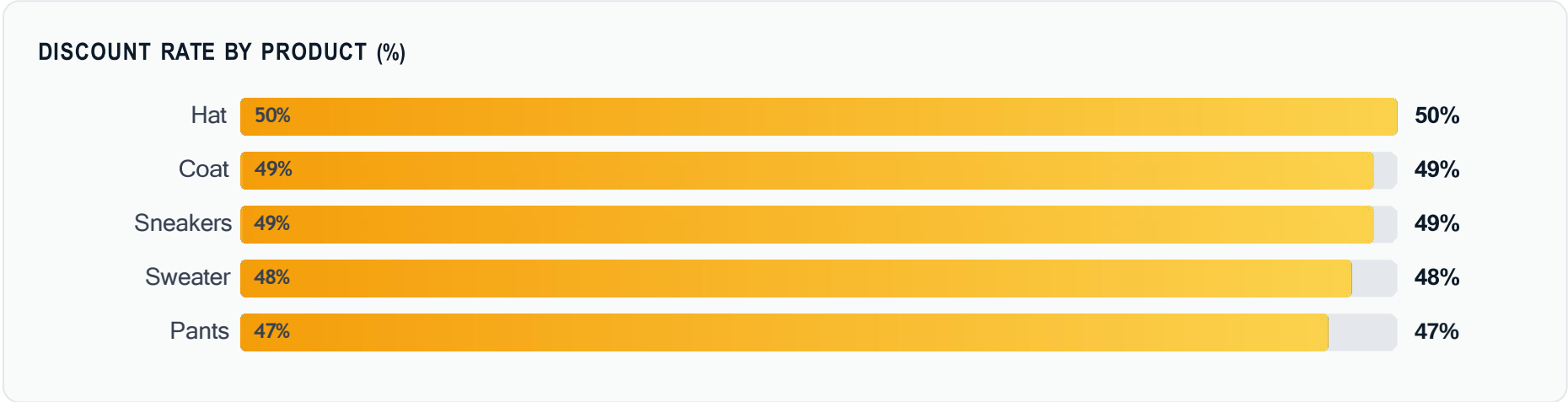


FINDING 5 — SUBSCRIPTION STATUS VS. SPENDING BEHAVIOUR

Subscription Status	No. of Customers	Total Revenue (USD)	Average Spend (USD)
Subscribed (Yes)	1,053	$62,645	$59
Not Subscribed (No)	2,847	$170,436	$59

⚠ Both cohorts record an identical average purchase of $59, conclusively demonstrating that subscription status has no statistically meaningful influence on per-transaction spend. This signals a critical gap in the subscription program's value proposition and incentive structure.

FINDING 6 — TOP 5 PRODUCTS BY DISCOUNT APPLICATION RATE



FINDING 7 — CUSTOMER SEGMENTATION (NEW / RETURNING / LOYAL)

Customer Segment	Definition (Previous Purchases)	Count	Share
Loyal	More than 10 previous purchases	3,116	79.9%
Returning	2 to 10 previous purchases	701	18.0%
New	Exactly 1 previous purchase	83	2.1%

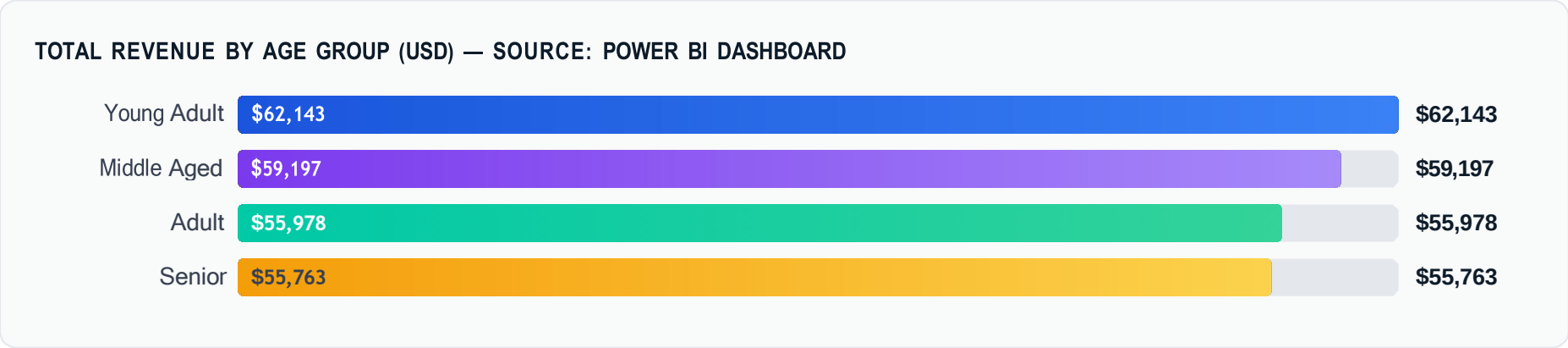
FINDING 8 — TOP 3 PRODUCTS BY CATEGORY

Rank	Category	Product	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

FINDING 9 — REPEAT BUYERS & SUBSCRIPTION LIKELIHOOD

Among customers with more than five previous purchases (repeat buyers), **958** hold an active subscription while **2,518** do not. Despite demonstrated loyalty through purchase history, the majority of repeat buyers remain outside the subscription program — a missed opportunity to formalize and monetize their engagement.

FINDING 10 — REVENUE CONTRIBUTION BY AGE GROUP



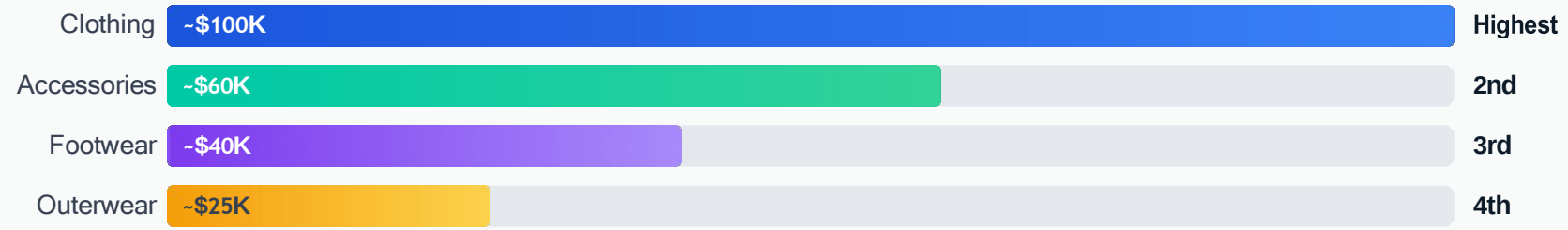
08 Dashboard Visualizations

An interactive Power BI Customer Behavior Dashboard was developed to enable real-time exploration of the findings. The dashboard incorporates dynamic slicers for Subscription Status, Gender, and Category, enabling stakeholders to drill into specific segments without requiring technical expertise. Key visual components reproduced below are derived from the Power BI dashboard output.

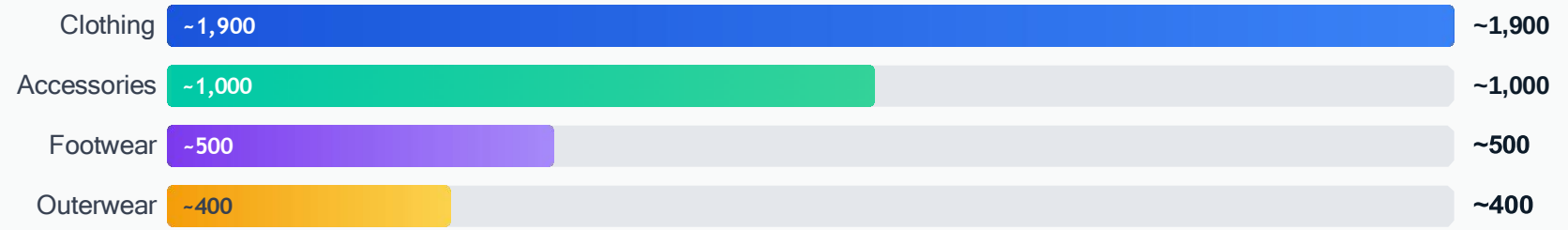
SUBSCRIPTION STATUS DISTRIBUTION

REVENUE & CUSTOMER COUNT BY CATEGORY

REVENUE BY CATEGORY — POWER BI BAR CHART (APPROX. VALUES)

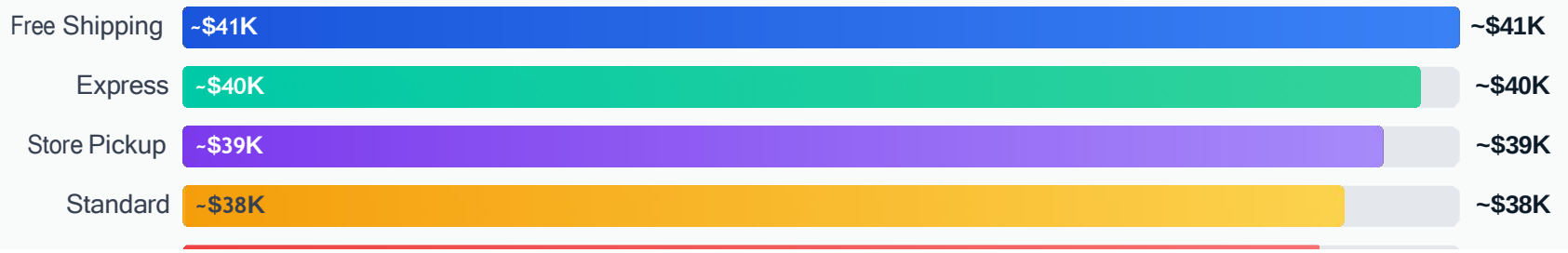


NUMBER OF CUSTOMERS BY CATEGORY — POWER BI BAR CHART



REVENUE BY SHIPPING TYPE

REVENUE BY SHIPPING METHOD — POWER BI AREA/BAR CHART (APPROX. FROM DASHBOARD)



09 Insights & Interpretation

STRATEGIC BUSINESS INSIGHTS

#	Insight	Supporting Evidence	Impact Level
1	Discounts are a net-positive lever — customers using discounts still exceed average spend	863 discount users spent above $59.76 avg	High
2	The subscription program is not influencing purchase behavior	Subscribed vs non-subscribed avg spend both $59	Critical
3	Young adults are the highest revenue-generating age segment	$62,143 total revenue — top of all age groups	High
4	Faster shipping options correlate with marginally higher average basket values	Express: $60 avg vs Standard: $58 avg	Medium
5	80% of the customer base is Loyal, yet subscription penetration is only 27%	3,116 Loyal customers; only 1,053 subscribers	Critical Gap
6	Clothing dominates both revenue and customer volume across all categories	Highest revenue (~$100K) and ~1,900 customers	High

#	Insight	Supporting Evidence	Impact Level
7	Hat leads in discount frequency (50%) — high promotional elasticity in accessories	Hat: 50% discount rate — highest of all products	Medium

TREND OBSERVATIONS

Revenue distribution across all four age groups is relatively balanced (ranging from $55,763 to $62,143), suggesting the product catalogue has broad demographic appeal. However, the slight revenue premium from young adults indicates an opportunity to double down on digital marketing channels that index heavily with this demographic. The near-uniform average spend across age groups ($59–$60) suggests pricing strategy is well-calibrated but that AOV (average order value) growth will require strategic interventions such as bundling or loyalty incentives rather than price adjustments alone.

10 Risks & Challenges

Risk	Category	Likelihood	Impact	Mitigation
3.7 Review Rating values imputed via median — may introduce category bias	Data Quality	Low	Medium	Validate imputed values against category distributions; flag in model output
Subscription program yields no spend uplift — risk of continued stagnation	Commercial	High	High	Conduct qualitative research; redesign subscription value proposition
Male revenue dominance may mask untapped female segment potential	Strategic	Medium	High	Segment female customers further; launch gender-targeted campaigns

Risk	Category	Likelihood	Impact	Mitigation
High loyal customer share (80%) not reflected in subscription uptake	Retention	High	High	Introduce loyalty-to-subscription conversion campaigns
Static dataset — analysis reflects historical behavior, not real-time trends	Technical	Medium	Medium	Establish automated data pipeline for continuous refresh of BI dashboards

11 Recommendations

HIGH

Revamp the Subscription Program – Introduce exclusive subscription benefits (early access, free express shipping, loyalty points) to differentiate subscribed vs. non-subscribed spend behavior. Current program shows zero measurable impact on AOV. *Owner: Marketing / Product Team*

HIGH

Target Loyal Non-Subscribers for Conversion – Launch a targeted campaign directed at the 2,518 repeat buyers who are not subscribed. Given their demonstrated purchase history, a personalized subscription offer with a trial incentive could significantly increase subscriber base. *Owner: CRM / Marketing*

HIGH

Expand Female Customer Engagement Strategy – With female revenue at $75,191 vs. male at $157,890, there is a clear opportunity to develop gender-tailored marketing campaigns, product recommendations, and category expansions to close the revenue gap. *Owner: Marketing / Merchandising*

MEDIUM

Scale Discount Strategy as an AOV Growth Tool – The data confirms that discounting drives higher basket values, not revenue dilution. Formalize a strategic discount calendar aligned to high-discount-rate products (Hat, Coat, Sneakers) to maximize revenue per transaction while managing margin thresholds. *Owner: Commercial / Pricing Team*

MEDIUM

Invest in Faster Fulfilment Options — Express and Free Shipping channels generate higher average revenue per order. Expanding same-day or next-day delivery capabilities, particularly in high-volume regions, is expected to positively impact both AOV and customer satisfaction. *Owner: Operations / Logistics*

LOW

Establish a Real-Time BI Pipeline — Transition from static dataset analysis to a continuous data pipeline feeding live Power BI dashboards. This will enable proactive decision-making and reduce analytical latency. *Owner: Data Engineering / IT*

12 Action Plan & Next Steps

#	Action Item	Owner	Timeline	Priority	Status
1	Redesign and relaunch subscription program with measurable spend incentives	Marketing	Q1 2026	High	Planned
2	Launch loyal-to-subscriber conversion campaign (CRM-driven)	CRM Team	Q1 2026	High	Planned
3	Develop female-targeted marketing strategy based on category analysis	Marketing	Q2 2026	High	Planned
4	Formalise discount calendar for high-discount-rate products	Commercial	Q1 2026	Medium	Planned
5	Evaluate logistics upgrade for same-day/next-day delivery expansion	Operations	Q2 2026	Medium	Planned
6	Build automated SQL-to-Power BI pipeline for real-time dashboard refresh	Data Engineering	Q3 2026	Low	Planned

13 Conclusion

This Customer Behavior Analysis successfully addressed all ten management business questions through a rigorous end-to-end data pipeline encompassing Python-based ETL processing, SQL Server analytical querying, and Power BI visual reporting. The analysis of 3,900 customer records yielded a comprehensive understanding of consumer purchasing behavior across gender, age, product category, shipping preference, and subscription status dimensions.

The most actionable finding of this engagement is the significant disconnect between customer loyalty and subscription program engagement. With 80% of the customer base classified as loyal, the organization possesses a robust foundation for subscription growth — provided the program is restructured to deliver tangible, measurable value that influences spending behavior. Similarly, the positive relationship between discounting and above-average basket values presents a clear opportunity for a structured promotional strategy that drives revenue rather than eroding it.

If the recommended actions are implemented systematically, the organization can anticipate meaningful improvements in subscription adoption rates, female customer revenue contribution, and average order values across all shipping channels — collectively positioning the business for sustainable, data-informed commercial growth.

> **Report Sign-Off:** Prepared and validated by Satish Aryalon **21 December 2025.** For queries or further analysis requests, contact the Business Intelligence team.

APPENDIX 14 – TECHNICAL REFERENCE

SQL Query Reference

All SQL queries executed against Microsoft SQL Server 2022 | Database: Customer_Behaviour | Table: customer

Q1 Total Revenue Generated by Male vs. Female Customers

```
SELECT  gender,
        SUM([purchase_amount_(usd)]) AS revenue_usd
FROM    customer
GROUP BY gender;
```

Result: Male – $157,890 | Female – $75,191

Q2 Customers Who Used a Discount but Spent Above the Average Purchase Amount

```
SELECT  customer_id,
        [purchase_amount_(usd)]
FROM    customer
WHERE   discount_applied = 'yes'
  AND [purchase_amount_(usd)] = (
          SELECT AVG([purchase_amount_(usd)])
```

```
            FROM   customer
        );
```

▸ *Result: 863 customers identified*

Q3 Top 5 Products with the Highest Average Review Rating

```
SELECT TOP 5
         item_purchased,
         ROUND(AVG(review_rating), 2) AS top_5_products_average_rating
FROM     customer
GROUP BY item_purchased
ORDER BY top_5_products_average_rating DESC;
```

▸ *Result: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Handbag (3.78)*

Q4 Average Purchase Amount — Standard vs. Express Shipping Comparison

```
SELECT   shipping_type,
         AVG([purchase_amount_(usd)]) AS average_amount
FROM     customer
WHERE    shipping_type IN ('standard', 'express')
GROUP BY shipping_type;
```

▸ *Result: Standard – $58 | Express – $60*

Q5 Subscribed vs. Non-Subscribed Customers — Spending Comparison

```sql
SELECT   subscription_status,
         COUNT(customer_id)           AS number_of_customer,
         SUM([purchase_amount_(usd)])   AS total_revenue,
         AVG([purchase_amount_(usd)])   AS average_revenue
FROM     customer
GROUP BY subscription_status;
```

- *Result: Yes – 1,053 customers | $62,645 total | $59 avg || No – 2,847 customers | $170,436 total | $59 avg*

Q6 Top 5 Products with Highest Percentage of Discount-Applied Purchases

```sql
SELECT TOP 5
         item_purchased,
         ROUND(
             100.0 * SUM(CASE WHEN discount_applied = 'yes'
                              THEN 1 ELSE 0 END)
             / COUNT(*), 2
         ) AS discount_rate
FROM     customer
GROUP BY item_purchased
ORDER BY discount_rate DESC;
```

- *Result: Hat (50%), Coat (49%), Sneakers (49%), Sweater (48%), Pants (47%)*

Q7 Customer Segmentation — New, Returning, and Loyal (CTE)

```sql
WITH customer_type AS (
    SELECT  customer_id,
            previous_purchases,
            CASE
```

```
                WHEN previous_purchases = 1              THEN 'new'
                WHEN previous_purchases BETWEEN 2 AND 10 THEN 'returning'
                ELSE                                          'loyal'
            END AS customer_segment
    FROM    customer
)
SELECT  customer_segment,
        COUNT(*) AS number_of_customer
FROM    customer_type
GROUP BY customer_segment;
```

▸ *Result: Loyal – 3,116 | Returning – 701 | New – 83*

Q8 Top 3 Most Purchased Products Within Each Category (Window Function)

```
WITH item_count AS (
    SELECT
        category,
        item_purchased,
        COUNT(customer_id)                          AS total_orders,
        ROW_NUMBER() OVER (
            PARTITION BY category
            ORDER BY     COUNT(customer_id)  DESC
        )                                               AS item_rank
    FROM    customer
    GROUP BY category, item_purchased
)
SELECT  item_rank, category, item_purchased, total_orders
FROM    item_count
WHERE   item_rank = 3;
```

▸ *Result: Accessories: Jewelry/Belt/Sunglasses | Clothing: Blouse/Pants/Shirt | Footwear: Sandals/Shoes/Sneakers | Outerwear: Jacket/Coat*

Q9 Are Repeat Buyers (5+ Previous Purchases) Likely to Subscribe?

```sql
SELECT   subscription_status,
         COUNT(customer_id) AS repeat_buyers
FROM     customer
WHERE    previous_purchases > 5
GROUP BY subscription_status;
```

▸ *Result: Subscribed – 958 repeat buyers | Not Subscribed – 2,518 repeat buyers*

Q10 Revenue Contribution by Age Group

```sql
SELECT   age_group,
         SUM([purchase_amount_(usd)]) AS total_revenue
FROM     customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

▸ *Result: young_adult – $62,143 | middle_aged – $59,197 | adult – $55,978 | senior – $55,763*

DB Database Connection & Data Export — Python (SQLAlchemy)

```python
import pyodbc
from sqlalchemy import create_engine

# Establish connection to Microsoft SQL Server 2022
engine = create_engine('mssql+pyodbc: /  .ODBC+Driver+17+for+SQL+Server')
```

```python
# Verify connection
pd.read_sql("SELECT @@VERSION AS version", engine)
# Output: Microsoft SQL Server 2022 (RTM) - 16.0.1000.6

# Export cleaned DataFrame to SQL Server
df.to_sql(
    name='customer',
    con=engine,
    schema='dbo',
    if_exists='replace',
    index=False
)
```

Database: Customer_Behaviour | Table: customer | Schema: dbo